

Total No. of Questions : 6]

SEAT No. :

P60

[Total No. of Pages : 2

TE/INSEM/APR-67
T.E. (I.T.) (Semester - II)
314451 : OPERATING SYSTEM
(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answers Q. 1 or Q.2, Q.3 or Q.4 Q.5 or Q.6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Aussme suitable data if necessary.*

Q1) a) Explain the following operating systems (i) Real time system (ii) Distributed system. **[6]**

b) What are the OS design consideration for multiprocessor and multicore architecture? **[4]**

OR

Q2) a) What is virtual machine? Explain the virtual machine architecture? **[6]**

b) What are the functions & services of OS? **[4]**

Q3) a) Draw and explain process state transition diagram. **[5]**

b) Define thread. Explain types of threads in details. **[5]**

OR

Q4) a) Consider following set of processes **[6]**

Process	AT	BT
P1	0	3
P2	1	5
P3	3	2
P4	9	5
P5	12	5

P.T.O.

Apply SJF preemptive & non preemptive calculate average waiting time & turn around time.

- b) Explain the concept of context switching with the help of neat diagram.[4]

Q5) a) Write and explain deadlock free solution for dining philosophers problem. [5]

- b) Enlist different ways to handle mutual exclusion. Give example of each.[5]

OR

Q6) a) Explain the necessary & sufficient condition for the occurrence of a deadlock. [6]

- b) Describe any two methods for deadlock recovery. [4]

